

ABC SENIOR SECONDARY SCHOOL

CBSE CLASS XII — BOARD EXAMINATION (THEORY)

Subject: Science / Mathematics

Time Allowed: 3 Hours | Maximum Marks: 80 | Date: _____

Name: _____ Roll No.: _____

Class/Section: _____

GENERAL INSTRUCTIONS

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Figures to the right indicate full marks.
4. Use of calculator is not permitted unless stated.

SECTION A — MULTIPLE CHOICE QUESTIONS

Each question carries 1 mark. Select the correct option.

Q1. (1 marks)

Read the following from your syllabus and answer the multiple-choice question.

Passage:

LIGHT 189 Exercises 1. Fill in the blanks: (a) An image that cannot be obtained on a screen is called _____. (b) Image formed by a convex _____ is always virtual and smaller in size. (c) An image formed by a _____ mirror is always of the same size as that of the object. (d) An image which can be obtained on a screen is called a _____ image. (e) An image formed by a concave...

Which statement is best supported by the passage above?

- (a) Fill in the blanks: (a) An image that cannot be obtained on a screen is called _____.
- (b) WINDS, STORMS AND CYCLONES 93 (iii) Amritsar (iv) Puri 9.
- (c) Which of the statements given below is correct?
- (d) (i) In winter the winds flow from the land to the ocean.

Q2. (1 marks)

Read the following from your syllabus and answer the multiple-choice question.

Passage:

poor conductor of heat and it has air trapped in between the fibres. Exercises 1. State similarities and differences between the laboratory thermometer and the clinical thermometer. 2. Give two examples each of conductors and insulators of heat. 3. Fill

in the blanks : (a) The hotness of an object is determined by its _____. (b) Temperature of boiling water cannot be measured by a...

Which statement is best supported by the passage above?

- (a) poor conductor of heat and it has air trapped in between the fibres.
- (b) (ii) In summer the winds flow from the land towards the ocean.
- (c) (iii) A cyclone is formed by a very high-pressure system with very high-speed winds revolving around it.
- (d) (iv) The coastline of India is not vulnerable to cyclones.

Q3. (1 marks)

Read the following from your syllabus and answer the multiple-choice question.

Passage:

A CIDS , B ASES AND S ALTS 53 Table 5.4 S. No. Test solution Initial colour Final colour 1. Shampoo (dilute solution) 2. Lemon juice 3. Soda water 4. Sodium hydrogencarbonate solution 5. Vinegar 6. Sugar solution 7. Common salt solution warm water. Keep the mixture for some time till water becomes coloured. Use the coloured water as an indicator. Add five drops of the indicator to each of the...

Which statement is best supported by the passage above?

- (a) Keep the mixture for some time till water becomes coloured.
- (b) Extended Learning — Activities and Projects 1.
- (c) You can perform the Activity 8.5 in the chapter slight
- (d) represents the positive terminal and the thicker, shorter line represents the negative terminal.

Q4. (1 marks)

Read the following from your syllabus and answer the multiple-choice question.

Passage:

LIGHT 189 Exercises 1. Fill in the blanks: (a) An image that cannot be obtained on a screen is called _____. (b) Image formed by a convex _____ is always virtual and smaller in size. (c) An image formed by a _____ mirror is always of the same size as that of the object. (d) An image which can be obtained on a screen is called a _____ image. (e) An image formed by a concave...

Which statement is best supported by the passage above?

- (a) Fill in the blanks: (a) An image that cannot be obtained on a screen is called _____.
- (b) For a switch the 'ON' position and the 'OFF' position are represented by the symbols as shown.
- (c) The wires used to connect the various components in a circuit are represented by lines.

(d) In Table 14.1, a battery and its symbol are also shown.

Q5. (1 marks)

Read the following from your syllabus and answer the multiple-choice question.

Passage:

fastest car. 12. shows the distance-time graph for the motion of two vehicles A and B.

Which one of them is moving faster? (i) (ii) Distance-time graph for the motion of two

cars 13. Which of the following distance-time graphs shows a truck moving with speed which is not constant? 2018-19

Which statement is best supported by the passage above?

Refer to the diagram below.

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(a) 13.15 shows the distance-time graph for the motion of two vehicles A and B.

(b) For some of the activities we may need more than one

(c) Consider the following statements: (a) Both acids and bases change colour of all indicators.

(d) (b) If an indicator gives a colour change with an acid, it does not give a change with a base.

Q6. (1 marks)

Read the following from your syllabus and answer the multiple-choice question.

Passage:

represents the positive terminal and the thicker, shorter line represents the negative terminal. For a switch the 'ON' position and the 'OFF' position are represented by the symbols as shown. The wires used to connect the various components in a circuit are represented by lines. In Table 14.1, a battery and its symbol are also shown. Do you know what a battery is? Look at the symbol of a...

Which statement is best supported by the passage above?

Refer to the diagram below.

(a) represents the positive terminal and the thicker, shorter line represents the negative terminal.

(b) (v) Motion of the hammer of an electric bell.

(c) WINDS, SEASONS AND CLIMATES 93 (iii) Amritsar (iv) Puri 9.

(d) Which of the statements given below is correct?

Q7. (1 marks)

Read the following from your syllabus and answer the multiple-choice question.

Passage:

LIGHT 189 Exercises 1. Fill in the blanks: (a) An image that cannot be obtained on a screen is called _____. (b) Image formed by a convex _____ is always virtual and smaller in size. (c) An image formed by a _____ mirror is always of the same size as that of the object. (d) An image which can be obtained on a screen is called a _____ image. (e) An image formed by a concave...

Which statement is best supported by the passage above?

(a) Fill in the blanks: (a) An image that cannot be obtained on a screen is called _____.

(b) WINDS, SEASONS AND CLIMATES 93 (iii) Amritsar (iv) Puri 9.

(c) Which of the statements given below is correct?

(d) (i) In winter the winds flow from the land to the ocean.

Q8. (1 marks)

Read the following from your syllabus and answer the multiple-choice question.

Passage:

poor conductor of heat and it has air trapped in between the fibres. Exercises 1. State similarities and differences between the laboratory thermometer and the clinical thermometer. 2. Give two examples each of conductors and insulators of heat. 3. Fill in the blanks : (a) The hotness of an object is determined by its _____. (b) Temperature of boiling water cannot be measured by a...

Which statement is best supported by the passage above?

- (a) poor conductor of heat and it has air trapped in between the fibres.
- (b) (ii) In summer the winds flow from the land towards the ocean.
- (c) (iii) A cyclone is formed by a very high-pressure system with very high-speed winds revolving around it.
- (d) (iv) The coastline of India is not vulnerable to cyclones.

SECTION B — VERY SHORT ANSWER

Answer in 40-60 words. Each question carries 2 marks.

Q9. (2 marks)

Answer in 40–60 words.

Explain or define the idea expressed in: “Find out the letters of English alphabet or any other language known to you in which the image formed in a plane mirror appears exactly like the letter itself.”

Context:

5. Find out the letters of English alphabet or any other language known to you in which the image formed in a plane mirror appears exactly like the letter itself. Discuss your findings. 2018-19

Q10. (2 marks)

Answer in 40–60 words.

Explain or define the idea expressed in: “(i) Yak (ii) Camel (iii) Goat (iv) Woolly dog 4.”

Context:

3. Which of the following does not yield wool? (i) Yak (ii) Camel (iii) Goat (iv) Woolly dog 4. What is meant by the following terms? (i) Rearing (ii) Shearing (iii) Sericulture 5. Given below is a sequence of steps in the processing of wool. Which are the missing steps? Add them. Shearing, _____, sorting,...

SECTION C — SHORT ANSWER

Answer in 60-80 words. Each question carries 3 marks.

Q11. (3 marks)

Answer in 60–100 words.

Summarise the main teaching points from this part of the chapter:

(i) roots (ii) stem (iii) flowers (iv) leaves 13. Why do farmers grow many fruits and vegetable crops inside large green houses? What are the advantages to the farmers? Extended Learning — Activities and Projects 1. Project Take a potted plant with broad leaves. Take two strips of black paper and cut out a small square in the centres. Cover a part of two leaves with these papers and secure...

Q12. (3 marks)

Answer in 60–100 words.

Summarise the main teaching points from this part of the chapter:

nature of the solution? Explain. 11. Consider the following statements: (a) Both acids and bases change colour of all indicators. (b) If an indicator gives a colour change with an acid, it does not give a change with a base. (c) If an indicator changes colour with a base, it does not change colour with an acid. (d) Change of colour in an acid and a base depends on the type of the indicator....

SECTION D — CASE BASED / SOURCE BASED

Read the passage and answer the sub-questions. Each question carries 4 marks.

Q13. (4 marks)

Read the following case based on two excerpts from the textbook.

Excerpt A:

of water in a pond or well rises. Which type of soil will allow water to reach a well faster and in greater amount? ■ Which type of soil retains the highest amount of water and which retains the least? ■ Can you suggest any method to let more rain water percolate and reach the water underground? 9.7 SOIL AND CROPS Different types of soils are found in different parts of India. In some parts...

Excerpt B:

different town. So, Paheli made a neat drawing showing how the various electric components were to be connected (). Setup to check how steady your hand is Can you draw this circuit conveniently? It made Boojho wonder if there was an easier way to represent these electric components. 14.1 SYMBOLS OF ELECTRIC COMPONENTS Some common electric components can be represented by...

Use the diagram if relevant.

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- (a) List two scientific facts or processes mentioned in Excerpt A. (2 marks)
- (b) How does Excerpt B add a new idea, example, or application compared to Excerpt A? (2 marks)

SECTION E — LONG ANSWER

Answer in detail with diagrams where necessary.

Q14. (5 marks)

Answer with clear introduction, body, and conclusion (use sub-headings).

Discuss the concepts illustrated below.

between her house and the school. 7. Show the shape of the distance-time graph for the motion in the following cases: (i) A car moving with a constant speed. (ii) A car parked on a side road. 8. Which of the following relations is correct? (i) $\text{Speed} = \text{Distance} \times \text{Time}$ (ii) $\text{Speed} = \frac{\text{Distance}}{\text{Time}}$ (iii) $\text{Speed} = \frac{\text{Time}}{\text{Distance}}$ (iv) $\text{Speed} = \frac{1}{\text{Distance} \times \text{Time}}$ 2018-19

SECTION F — EXTENDED RESPONSE

Answer with proper derivation / explanation. (8 marks)

Q15. (8 marks)

Answer with clear introduction, body, and conclusion (use sub-headings).

Discuss the concepts illustrated below.

examples given above, a higher speed seems to indicate that a given distance has been covered in a shorter time, or a larger distance covered in a given time. The most convenient way to find out which of the two or more objects is moving faster is to

compare the distances moved by them in a unit time. Thus, if we know the distance covered by two buses in one hour, we can tell which one is faster. We call the distance covered by an object in a unit time as the speed of the object.